Lab 4 – Linear Monopole Antenna

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Pre-Lab

This lab focuses on the design and analysis of a linear quarter-wave monopole antenna operating at 2.2 GHz. The system is modeled in HFSS to evaluate its input impedance, matching bandwidth, and radiation pattern. Performance is compared between an ideal monopole over an infinite ground plane and a realistic monopole over a finite ground plane, and the effect of inserting a coaxial quarter-wave transformer on these characteristics is investigated.

Obj 1

The wavelength used for design was derived directly from the design frequency of 2.2 GHz. This in turn was used to calculate the standard length of the quarter-wavelength monopole. The quarter-wavelength monopole has a standard input impedance of 36.5+j21.5 Ohms, and maximum directivity of 3.286, or 5.167 dB. The input reactance is inductive, indicating that the antenna should be shortened to reach resonance.

Obj-1)

$l = \frac{\lambda}{4} = \frac{c}{4f} = 34.1\,mm$ $\lambda = 0.136\,m$

$Z_{in} = 36.5 + j21.25$ ← Inductive: Shorten

$D_0 = 3.286 = 5.167\,dB$

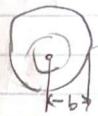
Obj 2

The rule of thumb for the size of a finite ground plane on a quarter wave monopole antenna is no less than a quarter-wavelength radius [1]

Obj 3

The radius of the inner coax core is designed to set the characteristic impedance to 50 Ω using PEC conductors and Roger RT/duroid 5880 dielectric.

Obs-3) $Z_0 = 50\,\Omega$



$b = 2.75$ mm $\quad \boxed{a = 0.79\text{ mm}}$

$\mu =$

for close to lossless TL:

$$Z_0 \approx \sqrt{\frac{L'}{C'}} \qquad Z_0^2 = \frac{\frac{\mu}{2\pi}\left[\ln(b/a)\right]^2}{2\pi\epsilon}$$

$$Z_0 = \sqrt{\frac{\mu}{\epsilon}}\,\frac{1}{2\pi}\left[\ln(b) - \ln(a)\right]$$

$\mu = 1\mu_0$

$\epsilon = 2.2\epsilon_0$

Lab

Part 1 — Quarter-Wave Monopole with Infinite Ground Plane

Model Setup

Figure 1 shows the model setup; a copper monopole with radius 1mm and length $\lambda/4$ surrounded by a vacuum cylinder. The bottom face of the cylinder is set as an infinite ground plane. A space of 0.005λ is left between the ground plane and the base of the antenna.

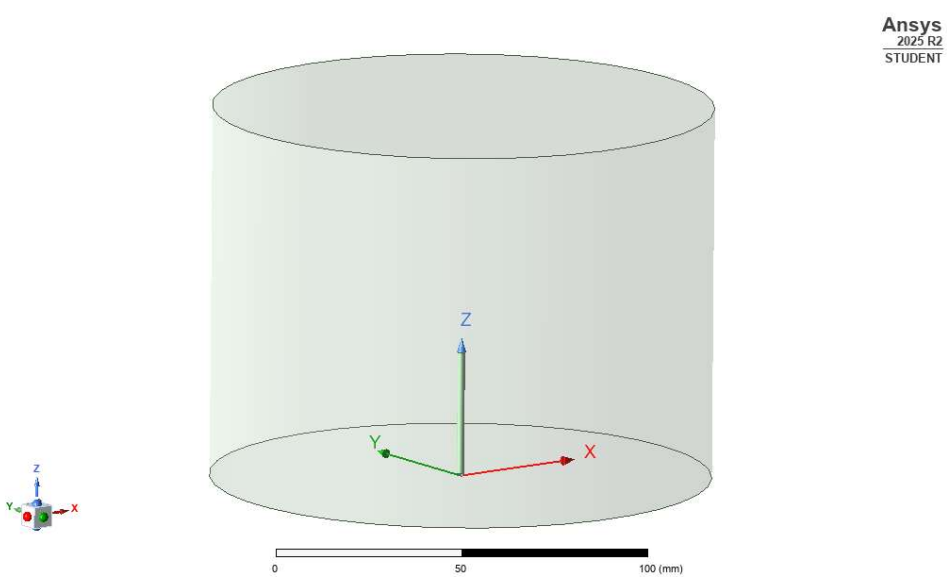


Figure 1: Device model for monopole antenna

Port Setup

Figure 2 shows the port setup for part 1. A single Driven Terminal, Lumped port connects the ground plane to the base of the monopole. The port is renormalized to 50 Ohms.

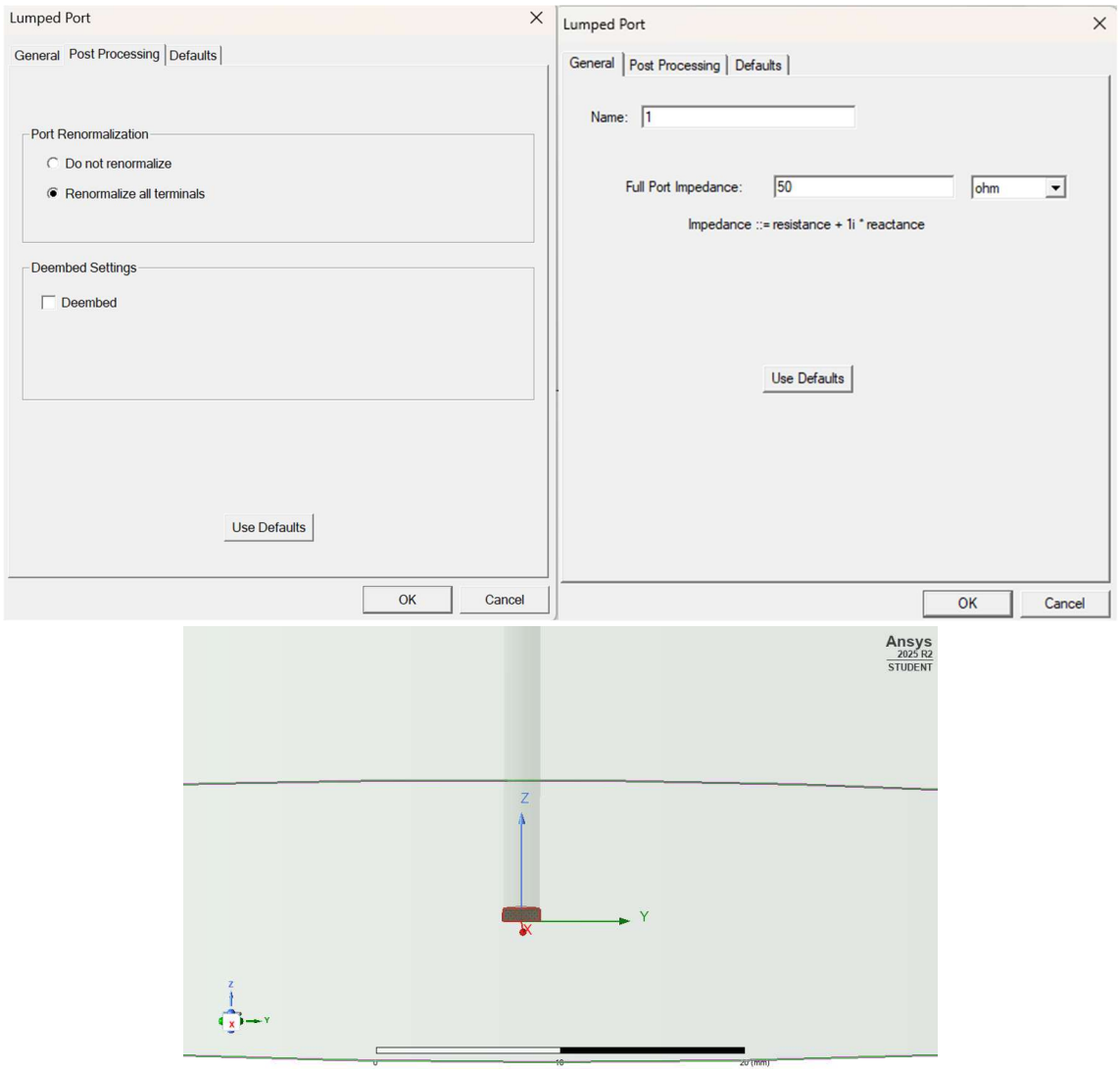


Figure 2: Port setup for monopole antenna

(Res 1) Figure 3 shows input impedance as a function of frequency between 1.1 GHz to 3.3 GHz the input impedance of the monopole at the design frequency is measured to be 60.32+j27.72. This result differs from the expected value of 36.5+j21.5, due to the use of a real copper conductor instead of an ideal PEC. However the input reactance is inductive as expected.

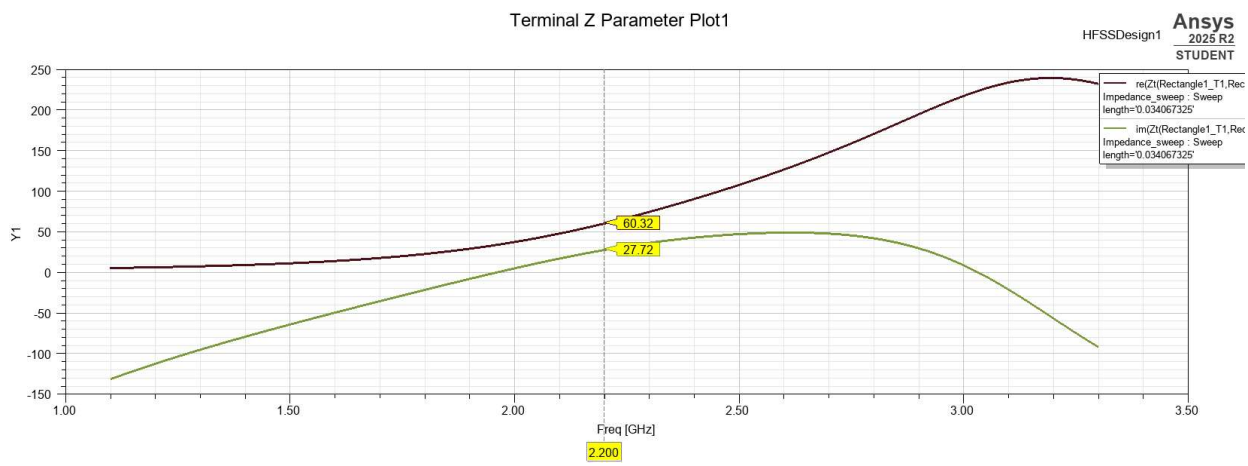


Figure 3: Input impedance sweep for infinite ground plane case

(Res 2) After shortening the antenna length to minimize the reactance magnitude, the monopole achieves resonance at the design frequency with an input resistance of 37.28 Ohms and a length of 30.5 cm. These results are displayed in Figure 4

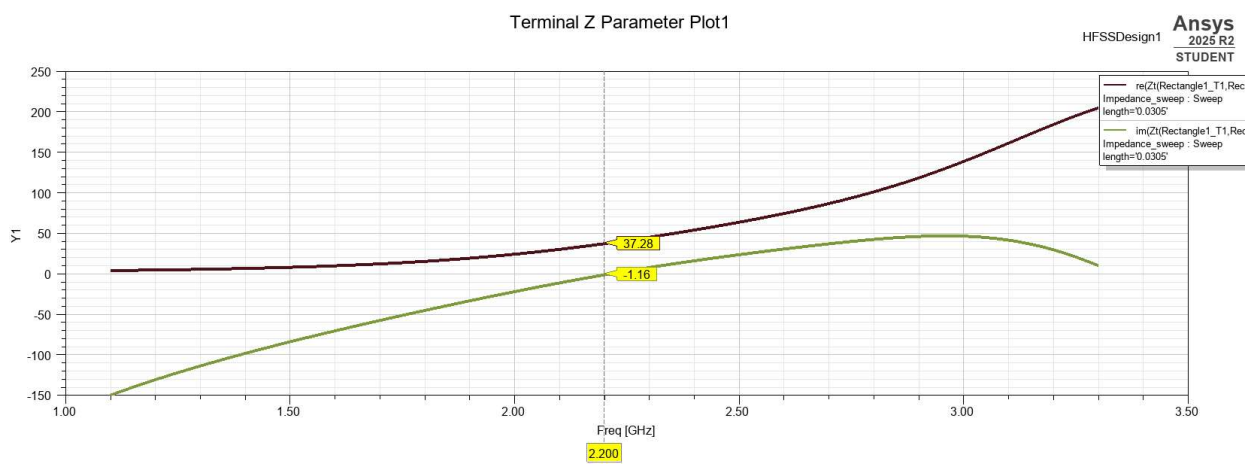


Figure 4: Input impedance sweep for monopole resonating at 2.2 GHz

(Res 3) The reflection coefficient and return loss at resonance were calculated using the simulated input impedance at 2.2 GHz as Z_L. *(Note: the hand calculations shown below use only the <u>resistive</u> component of the impedance. When the calculations were adjusted to use the full impedance, the return loss was 16.7 dB).*

Res 3) $\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} = -0.1494$ $RL = -20\log|\Gamma| = 16.5\,dB$

Shown in Figure 5, the simulated return loss at 2.2 GHz was 16.719 dB, closely matching expectations.

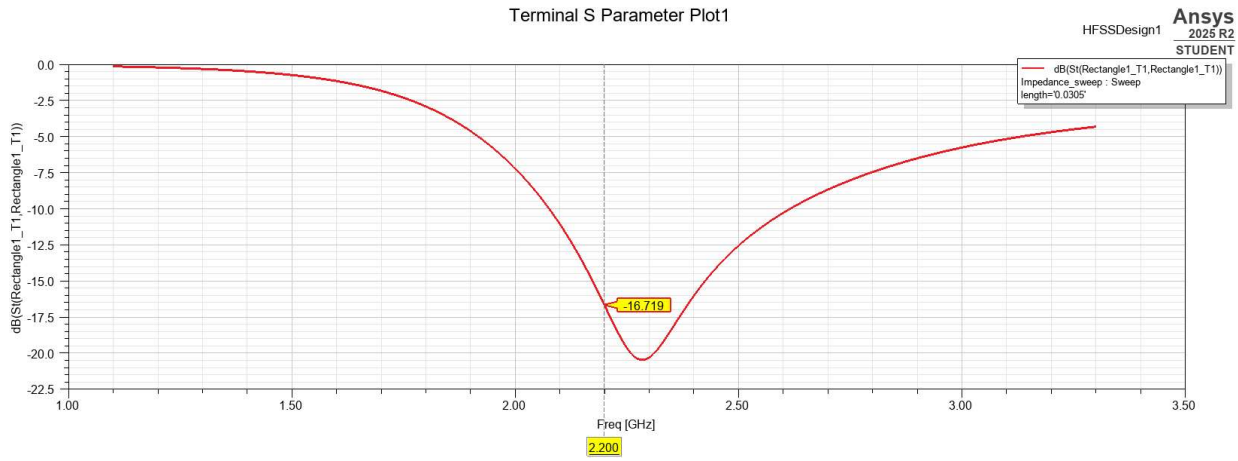


Figure 5: S(1,1) plot for resonant monopole showing negative return loss

(Res 4) Figures 6 and 7 show the plot of antenna directivity. Figure 6 shows the polar 3-D plot superimposed on the device model. Figure 7 plots the directivity as a function of theta, with phi set to 0°. The plot shows a maximum directivity of 3.1676 at $\theta = 90°$. This value is close to the expected value of 3.286 and the theoretical direction of maximum radiation for a monopole antenna. The discrepancy in D_0 is attributed to the change in length from the ideal quarter-wave monopole. As expected for a shorter antenna, the directivity is reduced.

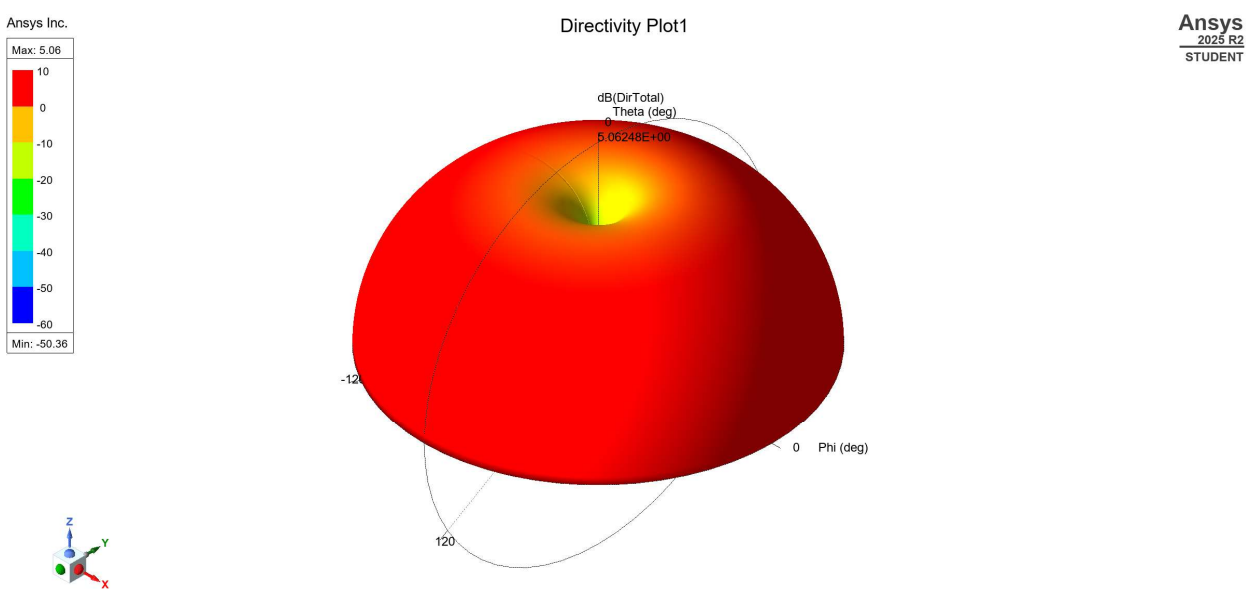


Figure 6: 3-D polar plot of antenna directivity (dB)

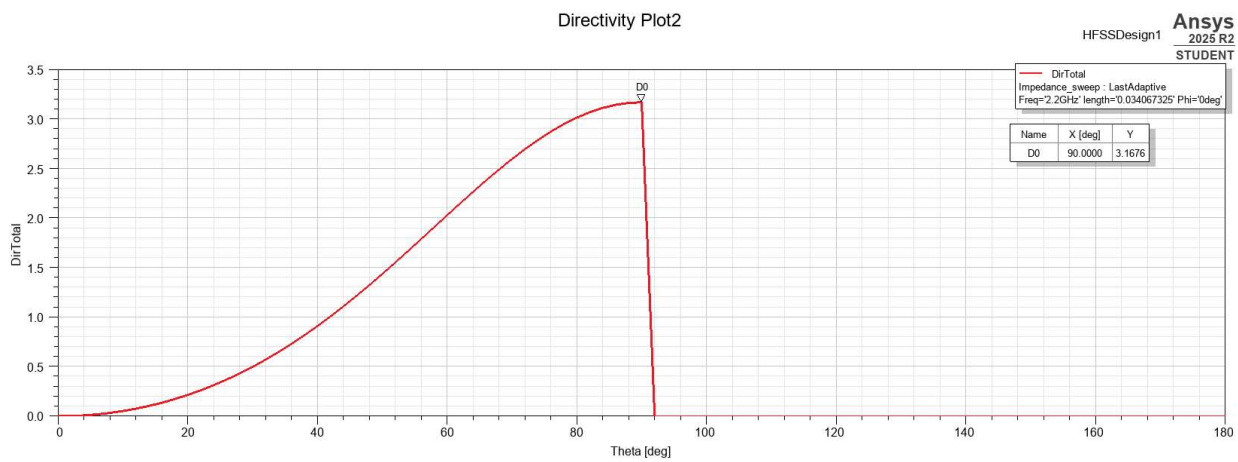


Figure 7: Rectangular plot of directivity vs. theta

Part 2 — Quarter-Wave Monopole with Finite Ground Plane

Model Setup

Figure 8 shows the model setup for part 2. The antenna geometry remains unchanged from part 1, but a finite copper ground plane with a radius of $\lambda/2$ and thickness 1mm is added at the base. The height of the air box is increased to keep a buffer of half a free-space wavelength between each end and the device geometry, and the radius is increased to a full free-space wavelength. The port setup remains unchanged from part 1.

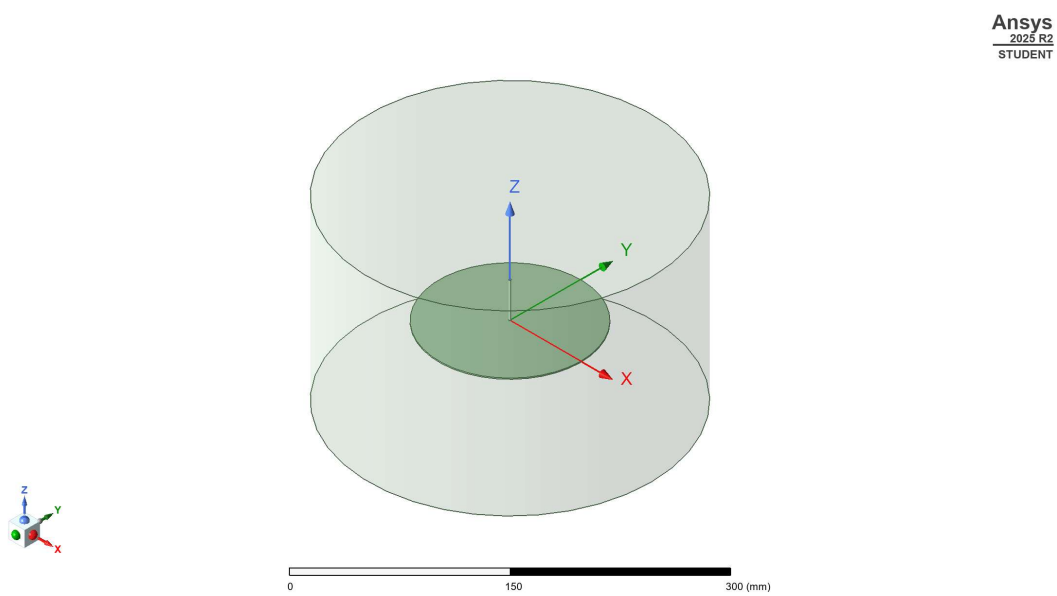


Figure 8: Device model for monopole antenna (finite ground)

(Res 5) Figure 9 shows input impedance as a function of frequency between 1.1 GHz to 3.3 GHz the input impedance of the monopole at the design frequency is measured to be 67.82+j30.82. The input reactance at design frequency is still inductive for the finite ground plane case.

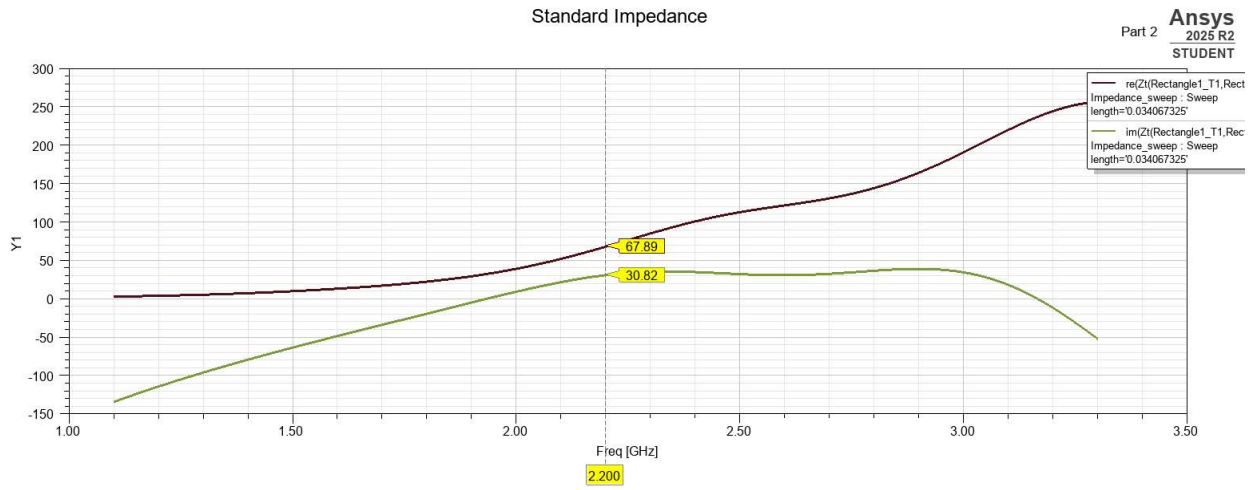


Figure 9: Input impedance sweep for finite ground plane case

(Res 6) After shortening the antenna length to minimize the reactance magnitude, the monopole achieves resonance at the design frequency with an input resistance of 41.19 Ohms and a length of 30.14 cm. These results are displayed in Figure 10.

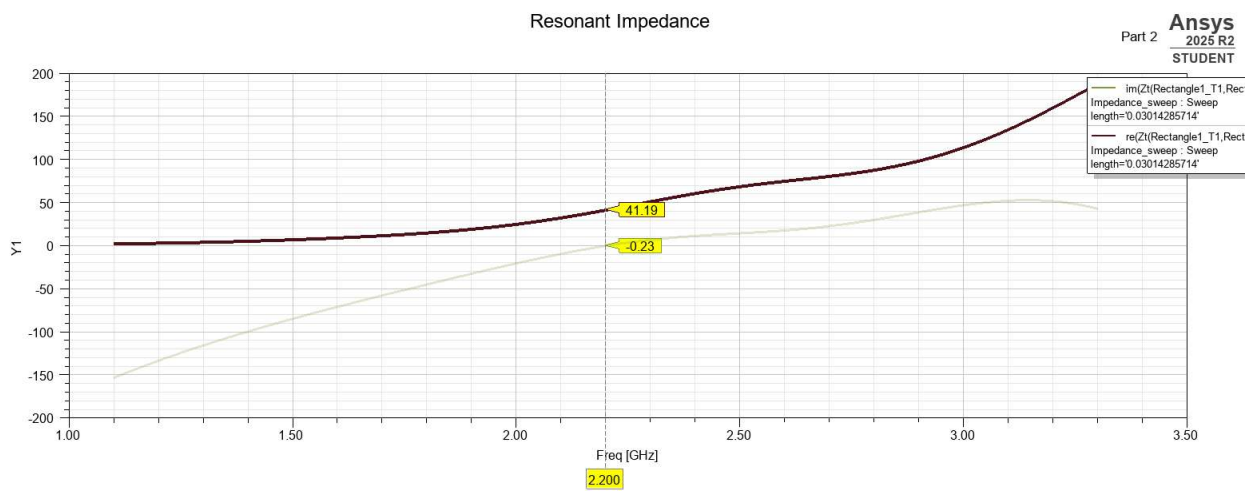


Figure 10: Input impedance sweep for monopole resonating at 2.2 GHz

(Res 7) The plot of scattering parameters shown in Figure 11 shows that return loss at design frequency was improved in the finite ground plane case. Although this contradicts the expected behavior, it matches the behavior of the resonance input resistances. Since the part 2 input resistance was closer to 50 Ohms, the matching was better which resulted in lower return loss.

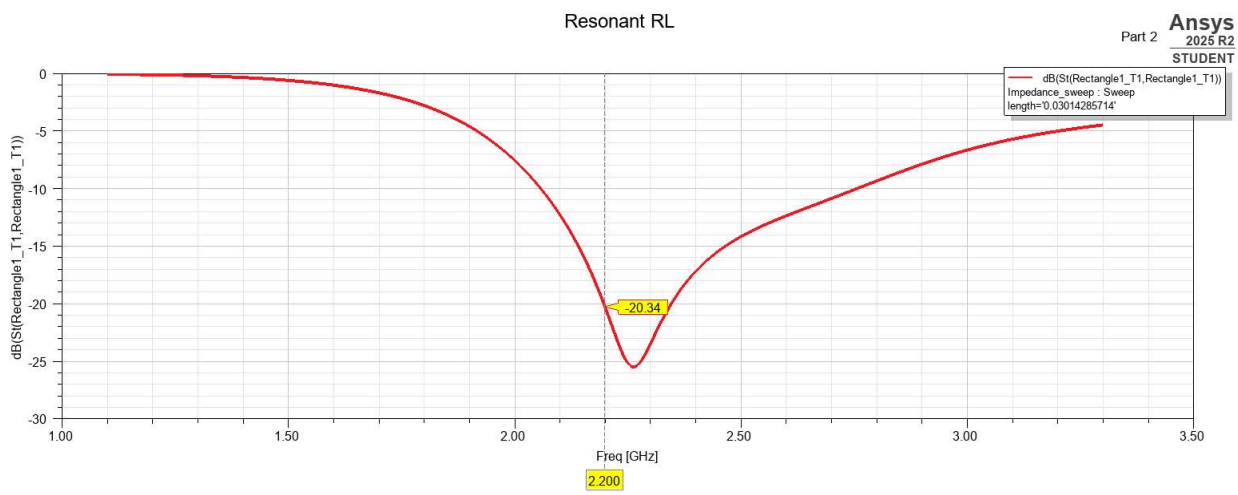


Figure 11: S(1,1) plot for resonant monopole showing negative return loss

(Res 8) Figure 12 shows the 3-D polar plot of antenna gain. Unlike the infinite case, it has non-zero gain for theta ≥ 90˚.

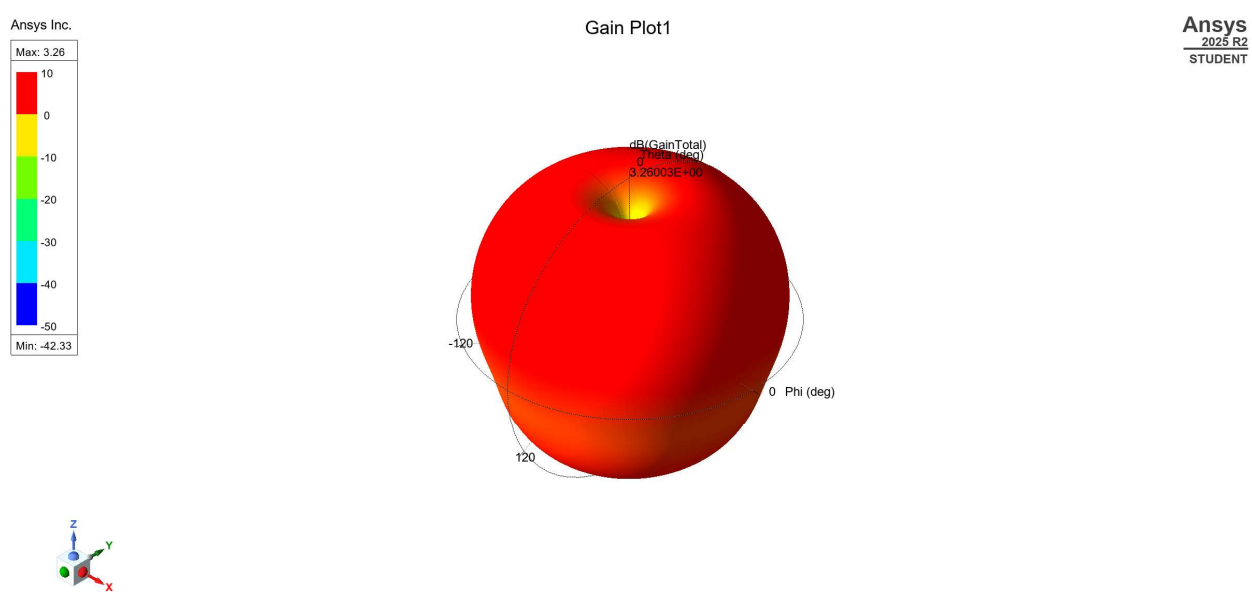


Figure 12: 3-D polar plot of antenna gain (dB)

Part 3 — Monopole with Matching network

(Res 9) Using the input impedance at resonance from part 2, a quarter-wave transformer section was designed to match the antenna impedance to the 50 Ohm feedline designed in the prelab.

The inner radius is set to 0.89 mm, with the length of the transformer section set to one quarter of the guided wavelength at 2.2 GHz.

Res 9) $Z_0 = 50\,\Omega$ $\quad b = 2.75\text{ mm}$ $\quad Z_L = 41.19\,\Omega$

$$Z_1 = \sqrt{Z_0 Z_L} = \sqrt{\frac{\mu}{\epsilon}}\,\frac{1}{2\pi}\left[\ln(b) - \ln(a)\right] \qquad \mu = \mu_0 \quad \epsilon = 2.2\epsilon_0$$

$$a = 0.89\text{ mm} \qquad \lambda_g = \frac{\lambda}{\sqrt{\epsilon_r}} \approx 92\text{ mm} \qquad \ell = \frac{\lambda_g}{4} = 23\text{ mm}$$

Model Setup

Figure 13 shows the model setup for part 3. The monopole and finite ground plane from part 2 are retained, but a 50-Ω coaxial feed line and a coaxial quarter-wave transformer section (length λg/4) are added below the ground plane, along with an SMA connector modeled as a hollow copper cylinder surrounding the coax. The air box is extended downward so that its bottom face is 0.5λ₀ below the 50-Ω line, while all other geometry parameters remain the same as in part 2.

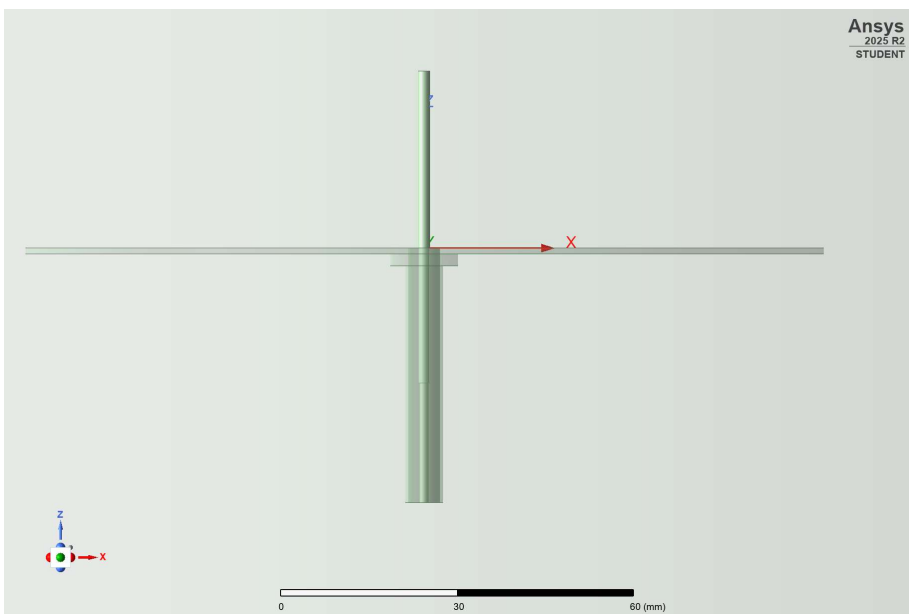
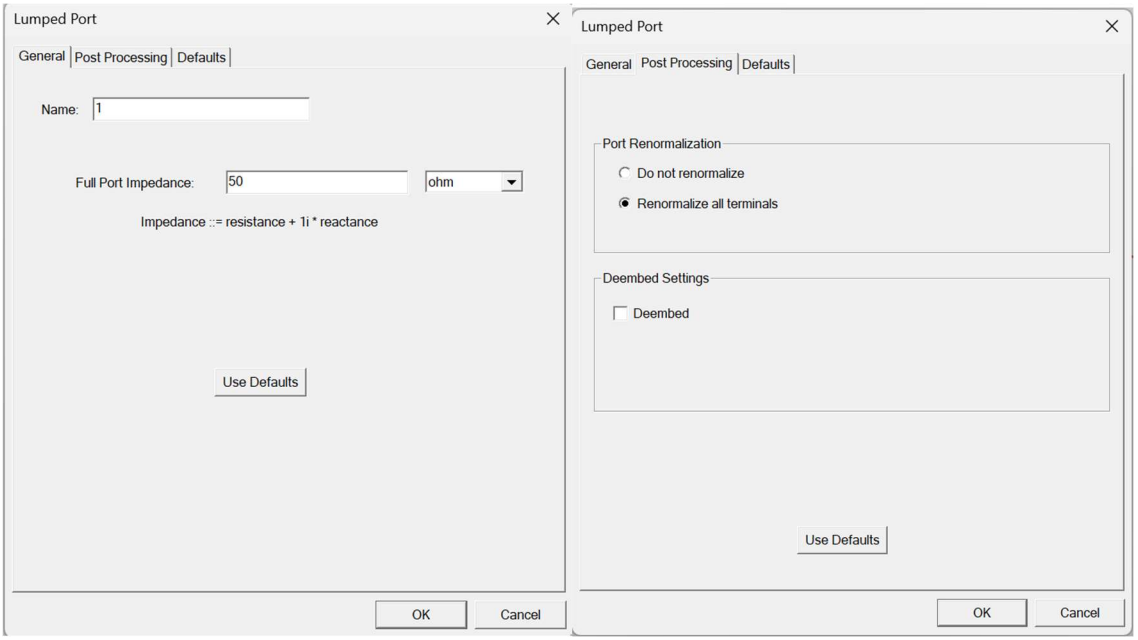


Figure 13: Device model for monopole antenna (with coax)

Port Setup

Figure 13 shows the port setup for part 3. The excitation is still a single Driven Terminal, Lumped port renormalized to 50~Ω, but it is now applied at the bottom of the coaxial feed line between the inner conductor and outer shell, so that the port drives the 50-Ω line, quarter-wave transformer, and monopole together.



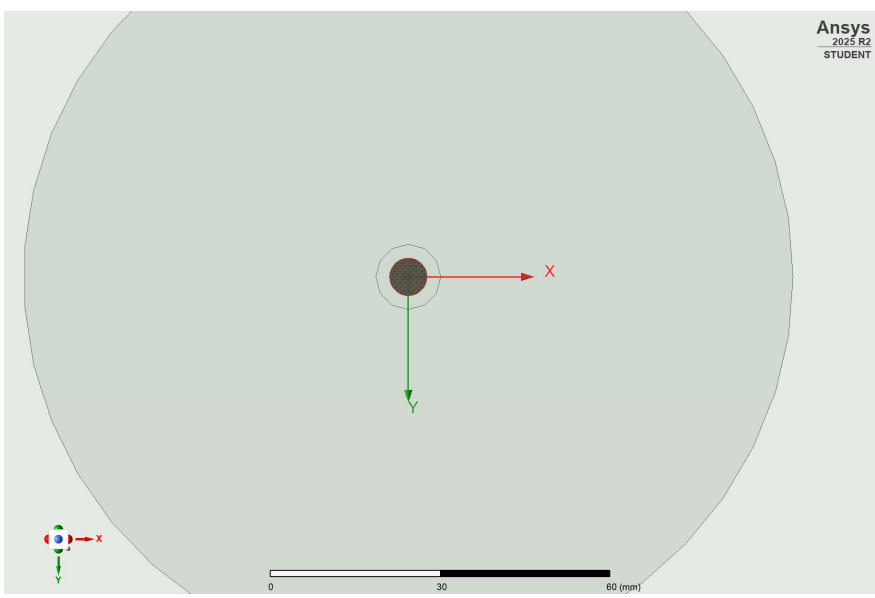


Figure 14: Port setup for monopole antenna (with coax)

(Res 10) The plot of scattering parameters shown in Figure 11 shows that return loss at design frequency was drastically improved by the matching network. At 2.2165 GHz, the center frequency is very close to the 2.2 GHz design frequency. The -10 dB matching bandwidth is 483 MHz.

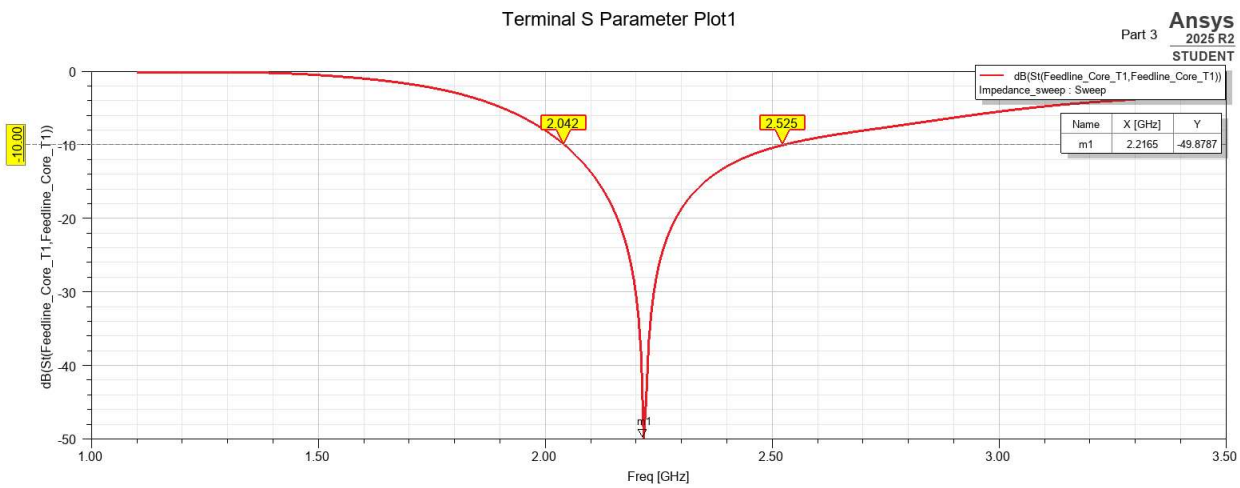


Figure 15: S(1,1) plot for monopole antenna with matching network

(Res 11) Figure 16 shows the frequency sweep of input impedance across the band 1.1-3.3 GHz. With the matching network, the impedance of the device is 48.99 – j2.76 at the design frequency. As expected, the real part is very close to the design value of 50 Ohms, and the imaginary part is close to zero.

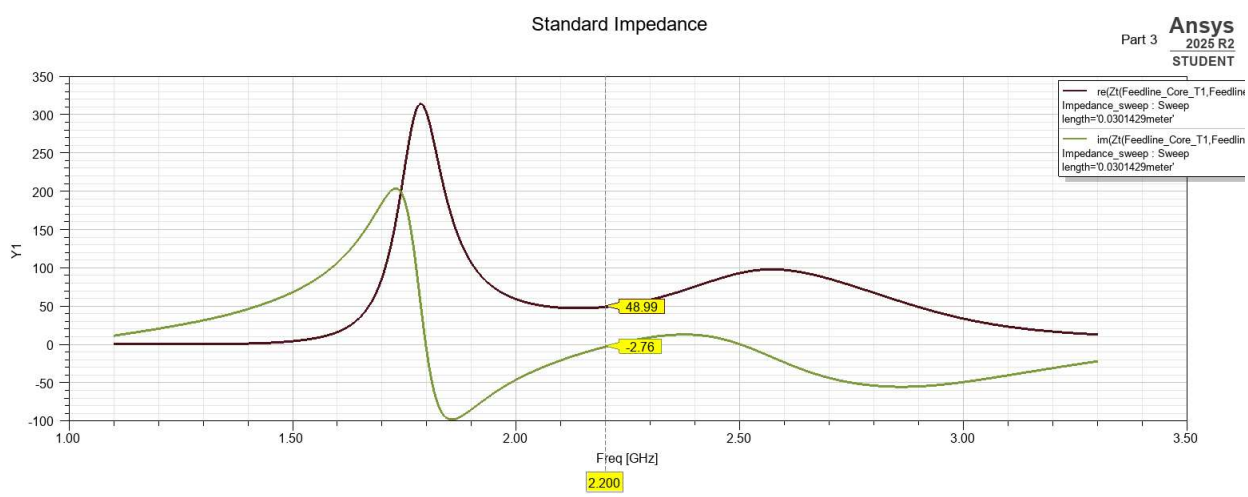


Figure 16: S(1,1) plot for monopole antenna with matching network

(Res 12) Figure 17 shows the 3-D polar plot of antenna gain. As in part 2, it has non-zero gain for theta ≥ 90˚.

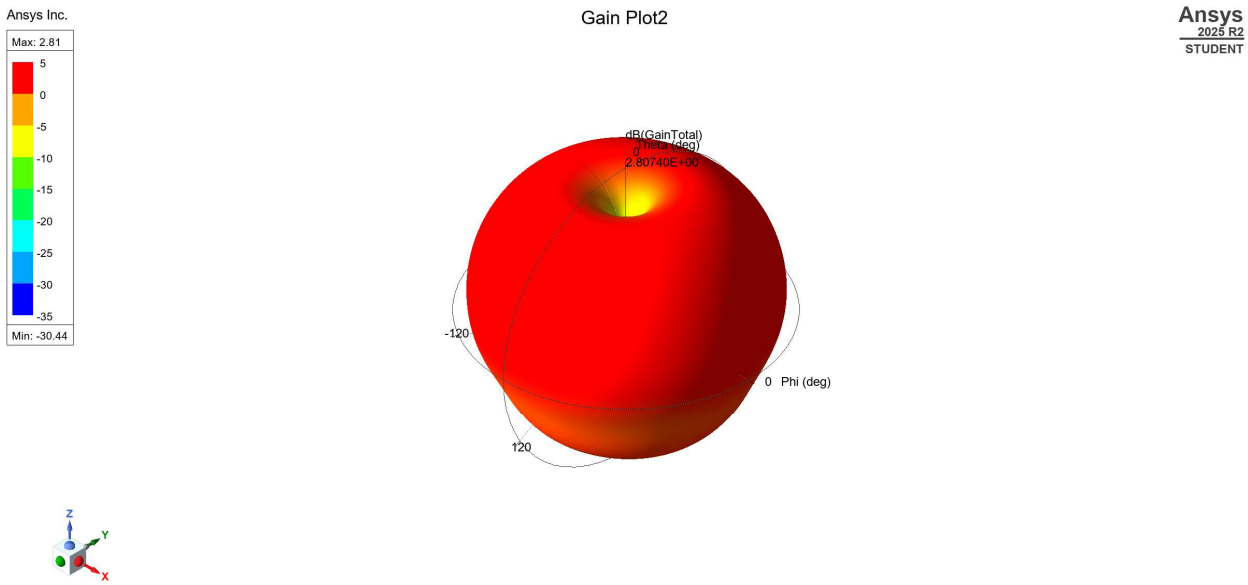


Figure 17: 3-D polar plot of antenna gain (dB)

References

[1] P. Gao, J. Li and W. Wang, "Study of Ground Plane Effects on Monopole Antenna Performance," *Electronics,* vol. 12, no. 12, p. 2681, 2023.

Report Checklist

1. I have included screenshots of all the HFSS models. **AM**
2. I have included screenshots of all port setups, including integration lines where appropriate. **AM**
3. I have included all necessary results in graphs, tables, or statements. **AM**
4. I have interpreted and commented on all my results. **AM**